香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

2025-2026 Term 1

Final Examination

December 19th, 2025; Time Allowed: 3 Hours

NAME (print) _____

CUHKSZ ID _____

Room No. _____

Seat No. _____

- **Show all your work**. Correct answers with little supporting work will not be given credit.
- Closed Book Exam: One piece of double-sided A4 reference paper, a scientific calculator, and a paper-based dictionary are allowed.
- Unless approved by the instructors, students who arrive more than 30 minutes late will NOT be admitted.
- The total points are 120 points. You need to finish ALL the questions in 3 hours (180 minutes).

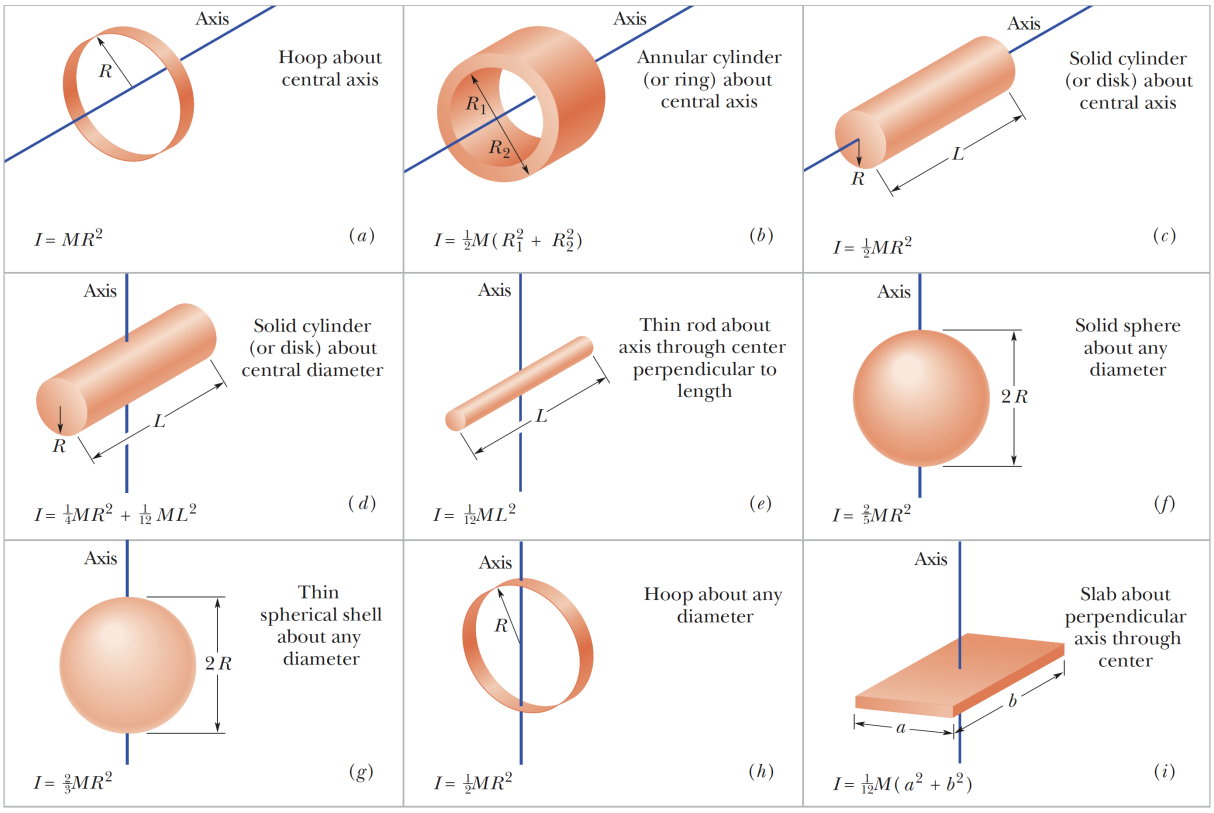
#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12	TOTAL
10	10	10	10	10	10	10	10	10	10	10	10	120

Summary of Basic Calculus:

$$
\begin{aligned}
&e^{ix} = \cos x + i \sin x, \\
&\frac{d}{dx} x^n = n x^{n-1}, \\
&\frac{d}{dx} e^{ax} = a e^{ax}, \\
&\frac{d}{dx} \ln ax = \frac{1}{x}, \\
&\frac{d}{dx}(uv) = v \frac{d}{dx} u + u \frac{d}{dx} v, \\
&\int x^n dx = \frac{x^{n+1}}{n+1} + C, \ \ (n \neq -1), \\
&\int \frac{dx}{x} = \ln x + C, \\
&\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad C \text{ is a constant.}
\end{aligned}
$$

Taylor series:

$$
\begin{aligned}
e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots \ (\text{all } x) \\
\sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + \frac{(-1)^n x^{2n+1}}{(2n+1)!} + \cdots \ (\text{all } x) \\
\cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + \frac{(-1)^n x^{2n}}{(2n)!} + \cdots \ (\text{all } x).
\end{aligned}
$$



1. The wave function for a traveling wave on a string is **(10 pts)**

$$y(x,t) = (1.50\,\text{mm})\sin(62.8\,\text{m}^{-1}x + 314\,\text{s}^{-1}t).$$

(a) In what direction does this wave travel? [Positive x / Negative x] direction **(1 pt)**

Solution: The wave travels in the $\boxed{\text{negative } x \text{ direction}}$. **(1 pt)**

(b) Write expressions for the transverse oscillation velocity $u(x,t)$ (or $v_y(x,t)$) and the transverse acceleration $a_y(x,t)$ of any point on the string. **(5 pts)**

Solution: Given:

$$A = 1.50 \times 10^{-3}\ \text{m}, \quad k = 62.8\ \text{m}^{-1}, \quad \omega = 314\ \text{s}^{-1}.$$

The transverse velocity is

$$u(x,t) = \frac{dy}{dt} \quad \textbf{(0.5 pt)}$$

$$u(x,t) = A\omega\cos(kx + \omega t)$$

$$u(x,t) = (1.50 \times 10^{-3}\ \text{m})(314\ \text{s}^{-1})\cos(62.8\ \text{m}^{-1}x + 314\ \text{s}^{-1}t) \quad \textbf{(1 pt)}$$

$$\boxed{u(x,t) = (4.71 \times 10^{-1}\ \text{m/s})\cos(62.8\ \text{m}^{-1}x + 314\ \text{s}^{-1}t)} \quad \textbf{(1 pt)}$$

The transverse acceleration is

$$a(x,t) = \frac{du}{dt} \quad \textbf{(0.5 pt)}$$

$$a(x,t) = -A\omega^2\sin(kx + \omega t)$$

$$a(x,t) = -(1.50 \times 10^{-3}\ \text{m})(314^2\ \text{s}^{-2})\sin(62.8\ \text{m}^{-1}x + 314\ \text{s}^{-1}t) \quad \textbf{(1 pt)}$$

$$\boxed{a(x,t) = -(1.48 \times 10^{2}\ \text{m/s}^2)\sin((62.8\ \text{m}^{-1}x + 314\ \text{s}^{-1}t)} \quad \textbf{(1 pt)}$$

(Please pay attention to the number of significant figures and the units.)

(c) If the tension in the string is 0.50 N, determine the linear mass density of the string. **(4 pts)**

Solution: The wave speed is

$$v = \frac{\omega}{k} = \frac{314\ \text{s}^{-1}}{62.8\ \text{m}^{-1}} = 5.00\ \text{m/s} \quad \textbf{(1 pt)}$$

For a stretched string:

$$v = \sqrt{\frac{T}{\mu}} \quad \Rightarrow \quad \mu = \frac{T}{v^2} \quad \textbf{(1 pt)}$$

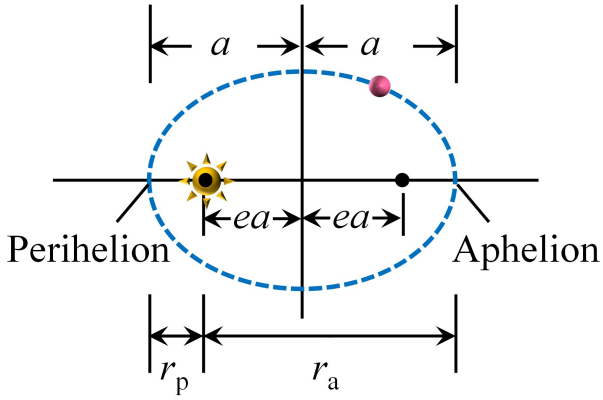
Thus,

$$\mu = \frac{0.50\ \text{N}}{(5.00\ \text{m/s})^2} = \frac{0.50\ \text{kg m/s}^2}{25.0\ \text{m}^2/\text{s}^2} = 0.0200\ \text{kg/m} \quad \textbf{(1 pt)}$$

$$\boxed{\mu = 2.00 \times 10^{-2}\ \text{kg/m}} \quad \textbf{(1 pt)}$$

(3 significant figures and the unit!)

2. The earth orbits the sun (mass m_s) in an elliptical orbit, as shown in the figure below. **(10 pts)**



(a) At its perihelion (closest point), the planet is at distance r_p from the sun. At its aphelion (farthest point), the planet is at distance r_a from the sun. Use Kepler's first law to express r_p and r_a in terms of the semi-major axis a and the eccentricity e. **(2 pts)**

Solution: $\boxed{r_p = a(1-e)}$ **(1 pt)**, $\boxed{r_a = a(1+e)}$ **(1 pt)**

(b) The planet's speed is v_p at perihelion, and v_a at aphelion. Use Kepler's second law to show that $v_p r_p = v_a r_a$. **(2 pts)**

Solution: Kepler's second law (equal area in equal time) implies conservation of angular momentum **(1 pt)**

$$\frac{1}{2} r_p v_p dt = \frac{1}{2} r_a v_a dt \quad \Rightarrow \quad v_p r_p = v_a r_a \quad \textbf{(1 pt)}$$

(c) Show that $v_p = \sqrt{\dfrac{Gm_s(1+e)}{a(1-e)}}$ and $v_a = \sqrt{\dfrac{Gm_s(1-e)}{a(1+e)}}$. **(4 pts)**

Solution: Total mechanical energy is conserved:

$$\frac{1}{2} m v_p^2 - \frac{Gm_s m}{r_p} = \frac{1}{2} m v_a^2 - \frac{Gmm}{r_a} \quad \textbf{(1 pt)}$$

Substitute $v_a = v_p \dfrac{r_p}{r_a}$ and solve for v_p **(1 pt)**

$$v_p = \sqrt{\frac{Gm_s(1+e)}{a(1-e)}} \quad \textbf{(1 pt)}, \quad v_a = \sqrt{\frac{Gm_s(1-e)}{a(1+e)}} \quad \textbf{(1 pt)}$$

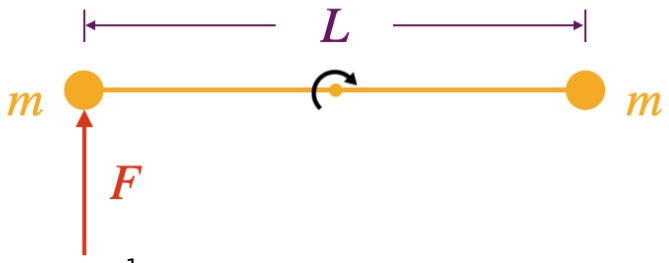
(d) Suppose a new planet is discovered orbiting the sun with a semi-major axis $a' = 2a$, where a is earth's semi-major axis. Use Kepler's third law to find the orbital period T' of this new planet in terms of earth's orbital period T_E. **(2 pts)**

Solution: Kepler's third law: $T^2 \propto a^3$. **(1 pt)**

For Earth: $T_E^2 = ka^3$, for new planet: $T'^2 = k(2a)^3 = 8ka^3$.

$$\Rightarrow \left(\frac{T'}{T_E}\right)^2 = 8 \quad \Rightarrow \quad \boxed{T' = T_E\sqrt{8} = 2\sqrt{2}\,T_E} \quad \textbf{(1 pt)}$$

3. A massless rigid rod has two identical spheres of mass $m = 2.0$ kg fixed at its ends, forming a dumbbell system on a frictionless table. The rod length is $L = 1.0$ m. The spheres can be considered as mass points. The system is free to rotate about a perpendicular axis through the center of the rod without friction. The system is initially at rest. **(10 pts)**



(a) Now, a force $F = 1.0 \times 10^1$ N is applied to the left sphere. The force is perpendicular to the rod and is upwards, see the figure. The system starts to rotate clockwise. Find the magnitude of angular acceleration α of the system. **(3 pts)**

Solution: $I_0 = 2mr^2 = 1.0\ \text{kg}\cdot\text{m}^2$ **(1 pt)**

$\tau = rF = 5.0\ \text{N}\cdot\text{m}$ **(1 pt)**

$\alpha = \dfrac{\tau}{I_0} = \boxed{5.0\ \text{rad/s}^2}$ **(1 pt)**

(b) The force F is applied for 2.0 s. During this period, the magnitude of the force remains constant, and its direction is always perpendicular to the rod, resulting in a constant torque about the rod center. Find the work W done by the torque during this period. Assume force F is the only force exerted on the system. (**Hint:** Use scientific notation to indicate significant figures.) **(2 pts)**

Solution: $\theta = \dfrac{1}{2}\alpha t^2 = 10\,\text{rad}$ **(1 pt)**

$W = \tau\theta = \boxed{5.0 \times 10^1\,\text{J}}$ 50 J is also correct. **(1 pt)**

It is also correct if using work-energy theorem $W = \Delta K$.

(c) After removing the force, the system rotates with a constant angular velocity ω_1. The two spheres are then pulled inwards by external forces pointing to the rod center along the rod, so that the distances between the rod center and each sphere decrease from 0.50 m to 0.25 m. The system now rotates with a constant angular velocity ω_2. Find the magnitude of angular velocity ω_2. (**Hint:** Use scientific notation to indicate significant figures.) **(3 pts)**

Solution: $I_f = 2mr_f^2 = 2 \times 2.0 \times (0.25)^2 = 0.25\ \text{kg}\cdot\text{m}^2$ **(1 pt)**

$\omega_i = \alpha t = 5.0\,\text{rad/s}$ **(0.5 pt)**

Angular momentum conservation: $I_i\omega_i = I_f\omega_f$ **(1 pt)**

$\omega_f = \boxed{4.0 \times 10^1\ \text{rad/s}}$ 40 rad/s is also correct. **(0.5 pt)**

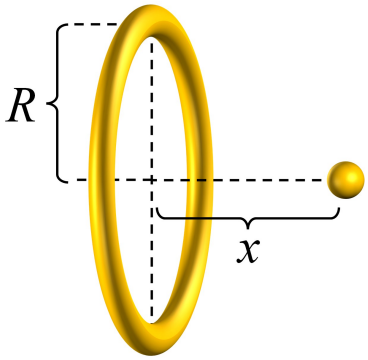
(d) What is the change in kinetic energy of the system before and after the contraction? (**Hint:** Use scientific notation to indicate significant figures.) **(2 pts)**

Solution: Before contraction: $K_i = \frac{1}{2}I_i\omega_i^2 = 5.0 \times 10^1\ \text{J}$ **(0.5 pt)**

After contraction: $K_f = \frac{1}{2}I_f\omega_f^2 = 2.0 \times 10^2\ \text{J}$ **(0.5 pt)**

Change in kinetic energy: $\Delta K = K_f - K_i = \boxed{1.5 \times 10^2\ \text{J}}$ 150 J is also correct. **(1 pt)**

4. A thin, uniform ring has radius R and mass M. A small uniform sphere of mass m is placed on the axis of the ring, at a distance x from its center. **(10 pts)**



(a) Calculate the gravitational potential energy U of the ring–sphere system. Take the potential energy to be zero when the ring and sphere are infinitely far apart. (**Hint:** Use the conservation of mechanical energy to determine the sign of U.) **(3 pts)**

Solution: Distance $r = \sqrt{R^2 + x^2}$ **(1 pt)**

The gravitational potential energy U of the system is:

$$\boxed{U = -\frac{GMm}{\sqrt{R^2 + x^2}}} \quad \textbf{(2 pts)}$$

- Correct expression for $\dfrac{GMm}{\sqrt{R^2 + x^2}}$ **(1 pt)**
- Negative sign **(1 pt)**

(b) Use $F(x) = -dU/dx$ to find the magnitude and direction of the gravitational force exerted on the sphere by the ring. **(2 pts)**

Solution: The gravitational force is **attractive**

$$\boxed{F(x) = -\frac{dU}{dx} = -\frac{GMmx}{(R^2 + x^2)^{3/2}}} \quad \textbf{(2 pts)}$$

- Correct derivative and expression **(1 pt)**
- Direction: negative sign implies attraction toward ring **(1 pt)**

(c) Derive your answers in (a) and (b) taking the limit $x \gg R$, and compare with the interaction between two point particles. **(2 pts)**

Solution: For $x \gg R$, approximate:

$$\boxed{U \approx -\frac{GMm}{x}} \quad \textbf{(1 pt)}, \quad \boxed{F(x) \approx -\frac{GMm}{x^2}} \quad \textbf{(1 pt)}$$

- Recognize point-mass limit.

(d) Show that for small displacements ($x \ll R$), the sphere undergoes simple harmonic motion. Derive its period T. **(3 pts)**

Solution: For $x \ll R$, the force is approximately: $F(x) \approx -\frac{GMm}{R^3}x$ **(1 pt)**

This matches Hooke's law $F = -kx$, so motion is **simple harmonic**.

Effective spring constant: $k = \frac{GMm}{R^3}$ **(1 pt)**

Period: $T = 2\pi\sqrt{\frac{m}{k}} = \boxed{2\pi\frac{R^{3/2}}{\sqrt{GM}}}$ **(1 pt)**

5. The atmospheric pressure on Earth decreases with increasing elevation y above sea level. Assume that air behaves as an ideal gas with constant molar mass M_m (Molar mass indicates the total mass of a substance per mole.), the temperature remains constant at T, and the gravitational acceleration is g. The pressure at sea level is p_0, and the ideal gas law $pV = nRT$ applies at all elevations. **(10 pts)**

(a) Derive the differential equation that relates the change in pressure dp to the change in elevation dy, noting that the air density decreases with increasing elevation. **(4 pts)**

Solution: We begin with the approximate linear relation for small height changes:

$$p = p_0 - \rho g y \quad \textbf{(1 pt)}$$

Differentiating both sides with respect to y gives

$$\frac{dp}{dy} = -\rho g. \quad \textbf{(1 pt)}$$

Since air density is not constant but decreases with height, we relate ρ to p using the ideal gas law:

$$pV = nRT, \qquad \rho = \frac{M_m n}{V},$$

Thus the density is

$$\rho = \frac{M_m}{RT} p \quad \textbf{(1 pt)}$$

Hence the governing differential equation is

$$\boxed{\frac{dp}{dy} = -\frac{M_m g}{RT} p} \quad \textbf{(1 pt)}$$

(b) Solve the equation above to obtain an expression for the atmospheric pressure $p(y)$ as a function of elevation y. **(6 pts)**

Solution: The equation is separable:

$$\frac{1}{p}\, dp = -\frac{M_m g}{RT}\, dy. \quad \textbf{(1.5 pt)}$$

Integrate from p_0 to $p(y)$ and from 0 to y:

$$\int_{p_0}^{p(y)} \frac{1}{p}\, dp = -\frac{M_m g}{RT} \int_0^y dy. \quad \textbf{(1.5 pt)}$$

Evaluate the integrals:

$$\ln p(y) - \ln p_0 = -\frac{M_m g}{RT} y. \quad \textbf{(1.5 pt)}$$

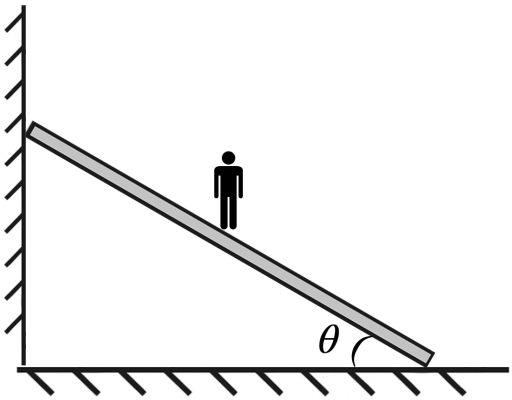
Exponentiate:

$$\frac{p(y)}{p_0} = \exp\left(-\frac{M_m g}{RT} y\right).$$

Thus the pressure as a function of elevation is

$$\boxed{p(y) = p_0 \exp\left(-\frac{M_m g}{RT} y\right)} \quad \textbf{(1.5 pt)}$$

6. A uniform ladder of length L and mass m rests on a rough horizontal floor and leans against a smooth vertical wall, making an angle $\theta = 30°$ with the floor. A person of mass m stands on the ladder. The coefficient of static friction between ladder and floor is $\mu_s = 1$. **(10 pts)**



(a) When the person stands at the midpoint of the ladder, determine the magnitude of friction at the bottom of the ladder. Will the ladder slip? [yes/no] **(4 pts)**

Solution: Equilibrium equations for forces:

$$\text{Horizontal: } f = N_{\text{wall}} \qquad \textbf{(0.5 pt)} \tag{1}$$

$$\text{Vertical: } N_{\text{floor}} = 2mg \qquad \textbf{(0.5 pt)} \tag{2}$$

Contact point on the floor as pivot. Equilibrium eqn. for torque:

$$N_{\text{wall}} L \sin\theta = mg\frac{L}{2}\cos\theta + mg\frac{L}{2}\cos\theta \qquad \textbf{(1 pt)} \tag{3}$$

From Eqs. (1) and (3): $f = N_{\text{wall}} = mg\cot\theta = \boxed{\sqrt{3}mg}$. **(0.5 pt)**

Maximum static friction: $f_{\max} = \mu_s N_{\text{floor}} = 2mg$. **(0.5 pt)**

Since $f = \sqrt{3}mg < f_{\max}$ **(0.5 pt)**, ladder does NOT slip. Answer is $\boxed{\text{No}}$ **(0.5 pt)**.

(b) If the person stands at the top of the ladder, will the ladder slip? [yes/no] You need to show steps. **(3 pts)**

Solution: Contact point on the floor as pivot. Torque:

$$N_{\text{wall}} L \sin\theta = mg\frac{L}{2}\cos\theta + mgL\cos\theta \qquad \textbf{(1 pt)}$$

$$N_{\text{wall}} = \frac{3\sqrt{3}}{2}mg \qquad \textbf{(0.5 pt)}$$

Friction force:

$$f_{\text{floor}} = N_{\text{wall}} = \frac{3\sqrt{3}}{2}mg \qquad \textbf{(0.5 pt)}$$

Since $f_{\text{floor}} = \frac{3\sqrt{3}}{2}mg > f_{\max} = 2mg$ **(0.5 pt)**, latter slips. Answer is $\boxed{\text{Yes}}$ **(0.5 pt)**.

(c) There is a critical angle θ_c at which the ladder is on the verge of slipping. The ladder will slip if $\theta < \theta_c$. When the person stands at a location of distance d to the bottom of the ladder (i.e., distance d is along the ladder), find $\tan\theta_c$. (**Hint:** You can use this result to verify your previous results.) **(3 pts)**

Solution: $N_{\text{wall}} L \sin\theta = mg\frac{L}{2}\cos\theta + mgd\cos\theta$ **(0.5 pt)**

$$N_{\text{wall}} = (\frac{d}{L} + \frac{1}{2})mg\cot\theta \qquad \textbf{(0.5 pt)}$$

Friction force:

$$f = N_{\text{wall}} = (\frac{d}{L} + \frac{1}{2})mg\cot\theta \qquad \textbf{(0.5 pt)}$$

If the ladder slips,

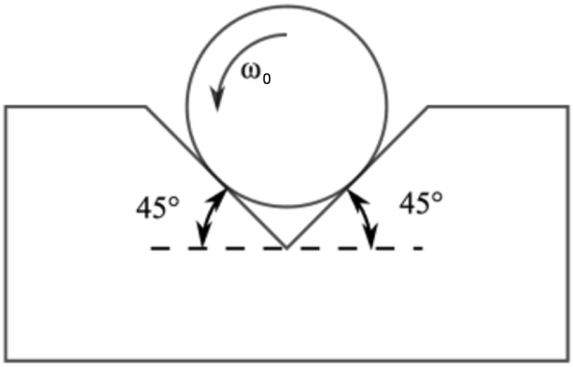
$$f \geq f_{\text{max}} \qquad \textbf{(0.5 pt)}$$

$$(\frac{d}{L} + \frac{1}{2})mg\cot\theta \geq 2mg \qquad \textbf{(0.5 pt)}$$

$$\tan\theta \leq \frac{1}{4} + \frac{d}{2L}$$

$\therefore \tan\theta_c = \boxed{\frac{1}{4} + \frac{d}{2L}}$ **(0.5 pt)**

7. A uniform solid cylinder of mass m and radius R is rotating counterclockwise in a V-shaped groove with an angular velocity ω_0. The normal force from the left surface on the cylinder is N_1, and from the right surface is N_2. Each surface makes an angle of 45° with the horizontal direction, as shown below. **(10 pts)**



(a) Assume there is no friction between the cylinder and the surfaces. Find the magnitudes of N_1 and N_2, respectively. **(4 pts)**

Solution:

Horizontal: $\frac{\sqrt{2}}{2}N_1 = \frac{\sqrt{2}}{2}N_2$ **(1 pt)**

Vertical: $\frac{\sqrt{2}}{2}N_1 + \frac{\sqrt{2}}{2}N_2 = mg$ **(1 pt)**

$N_1 = \boxed{\frac{\sqrt{2}}{2}mg}$ **(1 pt)**,

$N_2 = \boxed{\frac{\sqrt{2}}{2}mg}$ **(1 pt)**.

(b) Now assume that there is friction between the cylinder and the surface with coefficient of kinetic friction μ. As a consequence, the cylinder decelerates, but is still in contact with both surfaces. Find the angular acceleration α of the cylinder. You need to indicate its magnitude, and direction [into the page/out of the page] using right hand rule. **(6 pts)**

Solution: Horizontal: $\frac{\sqrt{2}}{2}N_1 = \frac{\sqrt{2}}{2}N_2 + \frac{\sqrt{2}}{2}\mu N_1 + \frac{\sqrt{2}}{2}\mu N_2$ **(1 pt)**

Vertical: $\frac{\sqrt{2}}{2}N_1 + \frac{\sqrt{2}}{2}N_2 + \frac{\sqrt{2}}{2}\mu N_1 = mg + \frac{\sqrt{2}}{2}\mu N_2$ **(1 pt)**

$N_1 = \frac{1+\mu}{1-\mu}N_2$

$N_1 = \frac{\sqrt{2}}{2}\frac{1+\mu}{1+\mu^2}mg$ **(1 pt)**,

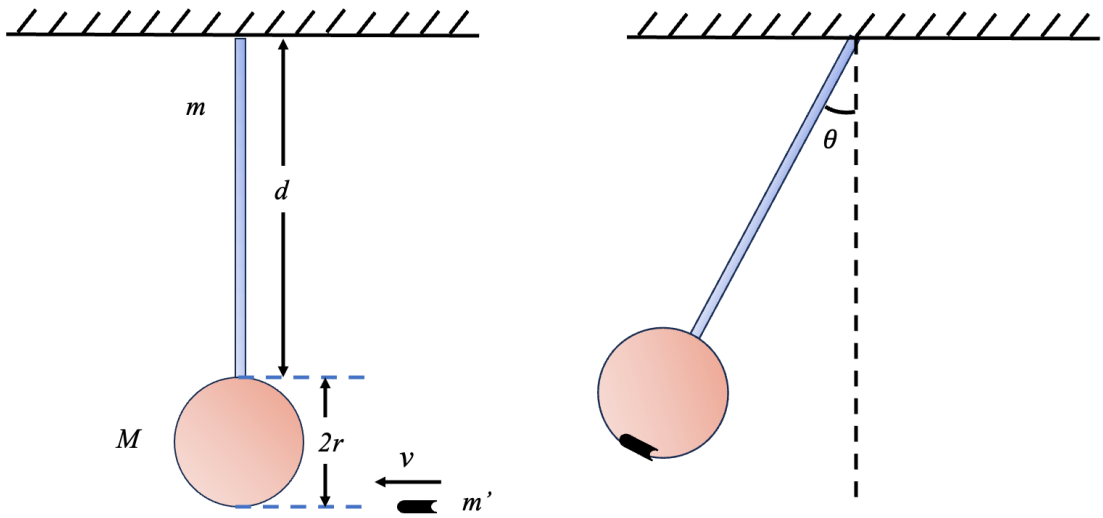
$N_2 = \frac{\sqrt{2}}{2}\frac{1-\mu}{1+\mu^2}mg$ **(1 pt)**.

$$\mu(N_1+N_2)R = \frac{1}{2}mR^2\alpha \qquad \textbf{(1 pt)}$$

$$\alpha = \boxed{\frac{2\sqrt{2}\mu}{1+\mu^2}\frac{g}{R}} \qquad \textbf{(0.5 pt)}$$

Since ω is out of the page, and ω decreases, direction of α is $\boxed{\text{into the page}}$. **(0.5 pt)**

8. In the figure below, a pendulum consists of a uniform disk with radius $r = 10.0\,\text{cm}$ and mass $M = 500\,\text{g}$ attached to a uniform thin rod with length $d = 500\,\text{mm}$ and mass $m = 250\,\text{g}$. The pendulum is initially at rest in the vertical position. A particle of mass $m' = 100\,\text{g}$ strikes the bottom rim of the disk with a horizontal velocity of $v = 2.00\,\text{m/s}$ and sticks to it. As a result, the pendulum begins to swing. Air resistance and friction at the pivot are negligible. Use $g = 9.80\,\text{m/s}^2$. Every digit in this question is a significant figure. **(10 pts)**



(a) Calculate the rotational inertial of the rod-disk-particle system about the pivot point after collision. **(3 pts)**

Solution:

$$I_{\text{rod}} = \frac{1}{12}md^2 + m\left(\frac{d}{2}\right)^2 = \frac{1}{3}md^2 = \frac{1}{3}(0.250)(0.500^2) = 0.0208\ \text{kg·m}^2 \quad \textbf{(1 pt)}$$

$$I_{\text{disk}} = \frac{1}{2}Mr^2 + M(d+r)^2 = \frac{1}{2}(0.500)(0.100^2) + 0.500(0.600^2) = 0.183\,\text{kg·m}^2 \quad \textbf{(1 pt)}$$

$$I_{\text{particle}} = m'(d+2r)^2 = 0.100(0.700^2) = 0.0490\ \text{kg·m}^2 \quad \textbf{(1 pt)}$$

$$I_{\text{total}} = 0.0208 + 0.1825 + 0.0490 = \boxed{0.253\ \text{kg·m}^2} = \boxed{2.53 \times 10^{-1}\,\text{kg·m}^2}$$

(Deduct 0.5 pt if the final answer is incorrect. And please pay attention to the significant figures.)

(b) Determine the angular speed of the entire system just after collision. **(2 pts)**

Solution: Angular momentum conservation:

$$L_{\text{initial}} = L_{\text{final}}.$$

$$L_{\text{initial}} = m'v(d+2r) = 0.100(2.00)(0.700) = 0.140\ \text{kg·m}^2/\text{s} \quad \textbf{(1 pt)}$$

$$L_{\text{final}} = L_{\text{initial}} = I_{\text{total}}\,\omega.$$

$$\omega = \frac{L_{\text{initial}}}{I_{\text{total}}} = \frac{0.140}{0.253} = \boxed{0.553\ \text{rad/s}} = \boxed{5.53 \times 10^{-1}\,\text{rad/s}} \quad \textbf{(1 pt)}$$

(c) What is the distance between the pivot point and the center of mass of rod-disk-particle system. **(1 pt)**

Solution: Write positions of each center of mass:

$$h_{\text{rod}} = \frac{d}{2} = 0.250 \text{ m}$$

$$h_{\text{disk}} = d + r = 0.500 + 0.100 = 0.600 \text{ m}$$

$$h_{\text{particle}} = d + 2r = 0.700 \text{ m}$$

Compute center of mass:

$$h_{\text{com}} = \frac{m_{\text{rod}}\, h_{\text{rod}} + M_{\text{disk}}\, h_{\text{disk}} + m'\, h_{\text{particle}}}{m_{\text{rod}} + M_{\text{disk}} + m'}.$$

$$h_{\text{com}} = \frac{(0.250)(0.250) + (0.500)(0.600) + (0.100)(0.700)}{0.850} = \boxed{0.509 \text{ m}} \quad \textbf{(1 pt)}$$

(d) Determine the maximum value of the angular displacement θ from the vertical equilibrium position attained by the rod-disk-particle pendulum after collision. **(2 pts)**

Solution: Kinetic energy right after collision:

$$K = \frac{1}{2} I_{\text{total}} \omega^2 = \frac{1}{2}(0.253)(0.553^2) = 0.0387 \text{ J} = 3.87 \times 10^{-2} \text{ J} \quad \textbf{(1 pt)}$$

Mechanical energy conservation:

$$\Delta U = K = M_{\text{tot}} g h_{\text{com}} (1 - \cos\theta) \quad \textbf{(1 pt)}$$

$$\cos\theta = \boxed{0.991} \quad \Rightarrow \quad \theta = \arccos(0.991) = \boxed{0.135 \text{ rad}} = \boxed{7.75°}$$

(Deduct 0.5 pt if the final answer is incorrect.)

(e) If the restoring torque is linearly related to the angular displacement: $\tau = -\kappa\theta$, the motion can be considered simple harmonic. Verify whether the swing motion described above is simple harmonic, and show the detailed reasoning. **(2 pts)**

Solution: For the an angular displacement θ from vertical, the gravitational restoring torque is:

$$\tau = -M_{\text{total}}\, g\, h_{\text{com}} \sin\theta \quad \textbf{(1 pt)}$$

For small angles after the collision

$$\theta_{\text{max}} \approx 0.135 \text{ rad} \quad \Rightarrow \quad \sin\theta \approx \theta \quad \textbf{(0.5 pt)}$$

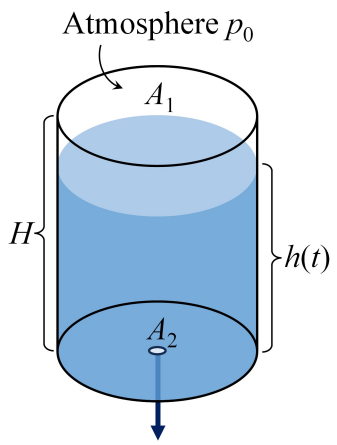
$$\tau \approx -M_{\text{total}}\, g\, h_{\text{com}}\, \theta$$

which matches the form

$$\tau = -\kappa\theta.$$

$$\boxed{\text{The swing motion is simple harmonic.}} \quad \textbf{(0.5 pt)}$$

9. A cylindrical tank of height H and uniform cross-sectional area A_1 is initially filled with water. The top of the tank is open to the atmosphere (p_0). At the bottom, there is a small orifice of area A_2, where $A_2 \ll A_1$. When the orifice is opened, water starts to flow out. Due to the large area ratio A_1/A_2, the water surface descends slowly, and the flow can be treated as quasi-steady, meaning that at any instant t, the continuity equation and Bernoulli's equation hold locally. **(10 pts)**



(a) Let $h(t)$ be the water height at time t, derive the instantaneous exit speed $v_2(t)$ in terms of g and $h(t)$. (**Hint:** Assume the rate of water level decrease is negligible.) **(2 pts)**

Solution: At point 1: pressure p_0, velocity ≈ 0, height $h(t)$;
At point 2: pressure p_0, velocity $v_2(t)$, height 0.

$$p_0 + \frac{1}{2}\rho(0)^2 + \rho g h(t) = p_0 + \frac{1}{2}\rho v_2^2 + \rho g(0) \quad \textbf{(1 pt)} \quad \Rightarrow \quad \boxed{v_2(t) = \sqrt{2gh(t)}} \quad \textbf{(1 pt)}$$

(b) Consider a small mass dm of water that drains from the tank during time dt. Show that the loss of gravitational potential energy of this dm equals its gain in kinetic energy. **(2 pts)**

Solution: Consider mass dm draining from height h to ground.
Gravitational potential energy lost: $dU = dm \cdot gh$ **(1 pt)**
Kinetic energy gained: $dK = \frac{1}{2} dm \cdot v_2^2 = \frac{1}{2} dm \cdot 2gh = dm \cdot gh = dU$ **(1 pt)**

(c) Find the rate of change of the height $\underline{dh/dt}$. Give your answer as a formula using A_1, A_2, g, and $h(t)$. (**Hint:** Use the result in (a) under the quasi-steady approximation.) **(3 pts)**

Solution: Volume flow rate: $A_2 v_2 = A_1 v_1$ **(1 pt)**, $v_1 = \left|\frac{dh}{dt}\right|$ **(1 pt)**

Substitute $v_2 = \sqrt{2gh(t)}$ $\Rightarrow$ $\boxed{\frac{dh}{dt} = -\frac{A_2}{A_1}\sqrt{2gh(t)}}$ **(1 pt)**

(d) Find the $\underline{\text{total time } T}$ that it takes for the water to completely flow out. Give your answer as a formula using A_1, A_2, H, and g. **(3 pts)**

Solution: Integrate $\frac{dh}{dt} = -\frac{A_2}{A_1}\sqrt{2gh}$ from $h = H$ to $h = 0$:

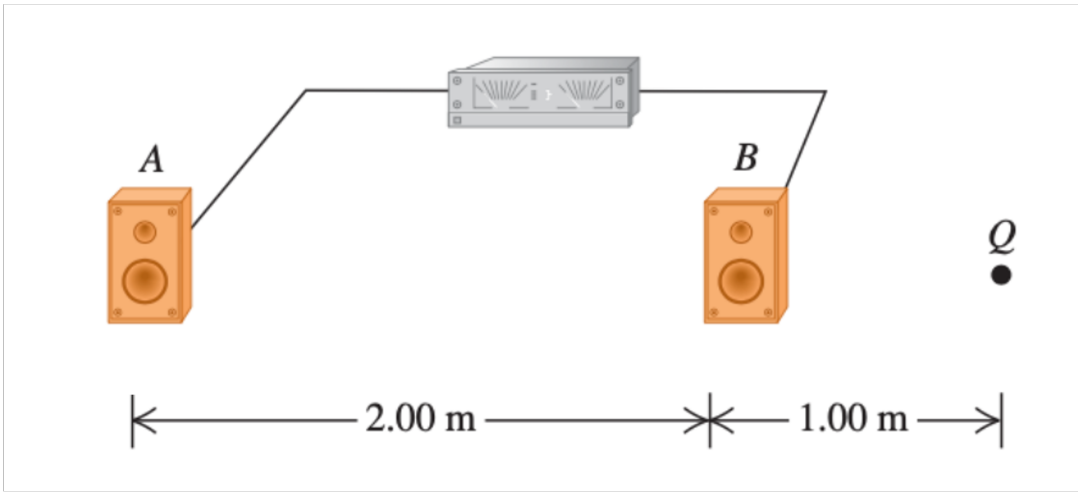
$$\int_H^0 h^{-1/2}\,dh = -\frac{A_2}{A_1}\sqrt{2g}\int_0^T dt \quad \textbf{(1 pt)} \quad \Rightarrow \quad 2\sqrt{H} = \frac{A_2}{A_1}\sqrt{2g}\,T \quad \textbf{(1 pt)}$$

$$\Rightarrow \quad \boxed{T = \frac{2A_1\sqrt{H}}{A_2\sqrt{2g}} = \frac{A_1}{A_2}\sqrt{\frac{2H}{g}}} \quad \textbf{(1pt)}$$

10. Two loudspeakers, A and B (Figure below), are driven by the same amplifier and emit identical sinusoidal waves that only have a phase difference of $\pi/3$. The wave functions are

$$S_A = A\cos(kx - \omega t), \qquad S_B = A\cos(kx - \omega t + \pi/3).$$

Speaker B is 2.00 m to the right of speaker A. Consider point Q along the extension of the line connecting the speakers, 1.00 m to the right of speaker B. Both speakers emit sound waves that travel directly from the speaker to point Q. The speed of sound in still air is 340 m/s. **(10 pts)**



(a) What is the lowest frequency for which fully constructive interference occurs at point Q? **(4 pts)**

Solution: Path difference:

$$\Delta L = d_A - d_B = 3.00\text{ m} - 1.00\text{ m} = 2.00\text{ m}$$

Because constructive interference requires

$$\left|\cos\left(\frac{\phi}{2}\right)\right| = 1 \quad \Rightarrow \quad \phi = 2n\pi, \quad \textbf{(1 pt)}$$

where the total phase difference is

$$\phi = \phi_0 + \frac{\Delta L}{\lambda}2\pi$$

Thus the constructive condition becomes

$$\frac{\pi}{3} + \frac{2\pi}{\lambda}(2.00\text{ m}) = 2n\pi \quad \textbf{(1 pt)}$$

Solve for wavelength:

$$\lambda = \frac{6\Delta L}{6n-1} \quad \textbf{(or any other format, 1 pt)}$$

The frequency is

$$f = \frac{v}{\lambda} = \frac{v(6n-1)}{6\Delta L}.$$

Choose the smallest integer giving a positive frequency: $n = 1$ **(0.5 pt)**

$$f = \frac{340\,(6\cdot 1 - 1)}{6\times 2.00} = \boxed{142\text{ Hz}} = \boxed{1.42\times 10^2\text{ Hz}}. \quad \textbf{(0.5 pt)}$$

(b) What is the second-lowest frequency for which fully destructive interference occurs at point Q? **(4 pts)**

Solution: Because destructive interference requires

$$\left|\cos\left(\frac{\phi}{2}\right)\right| = 0 \quad \Rightarrow \quad \phi = (2n+1)\pi, \quad \textbf{(1 pt)}$$

where the total phase difference is

$$\phi = \phi_0 + \frac{\Delta L}{\lambda} 2\pi$$

Thus the destructive condition becomes

$$\frac{\pi}{3} + \frac{2\pi}{\lambda}\Delta L = (2n+1)\pi \quad \textbf{(1 pt)}$$

Solving for the wavelength:

$$\lambda = \frac{6\Delta L}{6n+2} \quad \textbf{(or any other format, 1 pt)}$$

The frequency is

$$f = \frac{v}{\lambda} = \frac{v(6n+2)}{6\Delta L}$$

Choose the integer giving a second-lowest positive frequency: $n = 1$ **(0.5 pt)**

$$f == \frac{v(6n+2)}{6\Delta L} = \frac{340(6+2)}{6(2.00)} = \boxed{227\text{ Hz}} = \boxed{2.27 \times 10^2\text{ Hz}}. \quad \textbf{(0.5 pt)}$$

(c) If speaker A is turned off and speaker B emits a 500 Hz tone, what frequency is heard by a person at Q who moves directly away from B at 2.00 m/s? **(2 pts)**

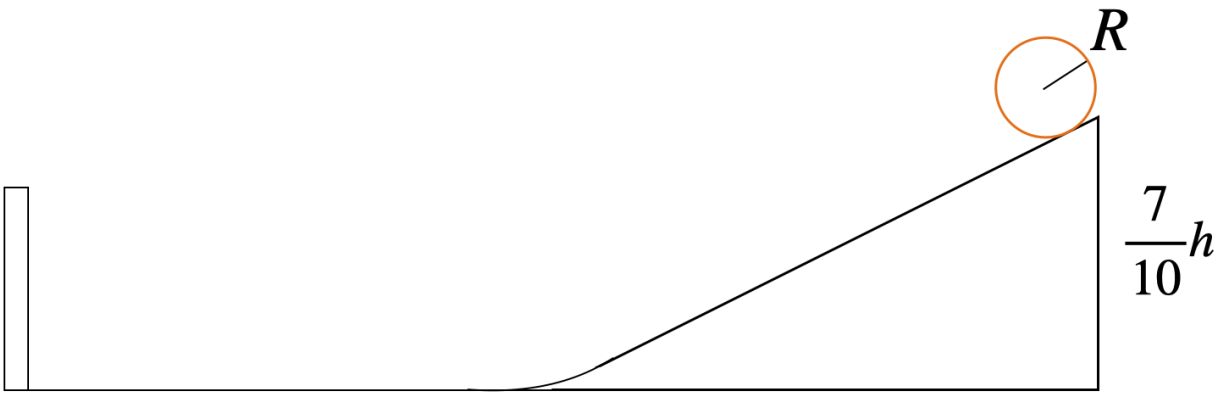
Solution: For a stationary source and a receiver moving away from it, the observed frequency is

$$f' = f\left(\frac{v - v_d}{v}\right) \quad \textbf{(1 pt)}$$

Substitute numerical values:

$$f' = 500\text{ Hz}\left(\frac{340\text{ m/s} - 2.00\text{ m/s}}{340\text{ m/s}}\right) = \boxed{497\text{ Hz}} \quad \textbf{(1 pt)}$$

11. A uniform solid sphere of mass m and radius R starts from rest and rolls without slipping down a rough slope of height $\frac{7}{10}h$, see the figure below. The acceleration due to gravity is g. **(10 pts)**



(a) When the sphere reaches the bottom of the slope, find the speed v of the center-of-mass of sphere. (**Hint:** center-of-mass of the sphere descends by $\frac{7}{10}h$ when it reaches the bottom.) **(3 pts)**

Solution:

$$I = \frac{2}{5}mR^2$$

Energy conservation: $mg\frac{7}{10}h = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ **(1 pt)**

Rolling without slipping: $v = R\omega$ **(1 pt)**

$$mg\frac{7}{10}h = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mR^2\right)\left(\frac{v}{R}\right)^2$$

$$v = \boxed{\sqrt{gh}}$$ **(1 pt)**

(b) The sphere then rolls without slipping to the left on a rough horizontal plane with v, until it undergoes an elastic collision with a frictionless wall at the left end. Immediately after the collision, the center-of-mass velocity of the sphere is equal in magnitude but opposite in direction . The sphere rolls with slipping to the right for a duration of t, before rolling without slipping again, and the sphere is still on the horizontal plane. Find the time t that it takes from collision to rolling without slipping again. The coefficient of kinetic friction between the sphere and the horizontal plane is μ. **(4 pts)**

Solution: Set into the page as the positive direction for angular velocity, and to the right as the positive direction for velocity. Right after collision, velocity is v **(0.5 pt)**, while the angular velocity is $-v/R$ **(0.5 pt)** since it is still out of the page as before collision.

The friction is $f = \mu mg$, therefore for translation:

Newton 2nd Law: $-\mu mg = ma$ (4)

$a = -\mu g$ **(0.5 pt)** (5)

Velocity at time t after collision: $v = v - \mu gt$ **(0.5 pt)** (6)

For rotation:

$$\mu mgR = \frac{2}{5}mR^2\alpha \rightarrow \alpha = \frac{5}{2}\frac{\mu g}{R} \qquad \textbf{(0.5 pt)}$$

$$\omega = -\frac{v}{R} + \frac{5}{2}\frac{\mu g}{R}t \qquad \textbf{(0.5 pt)}$$

To achieve rolling without slipping, we have

$$v = \omega R \rightarrow v - \mu gt = -v + \frac{5}{2}\mu gt \qquad \textbf{(0.5 pt)}$$

$$t = \frac{4v}{7\mu g} = \boxed{\frac{4}{7\mu}\sqrt{\frac{h}{g}}} \qquad \textbf{(0.5 pt)}$$

(c) From now on, assume that the vertical wall is rough. The kinetic friction coefficient between the sphere and the vertical wall is also μ, and $\mu < 0.2$. For the collision, we make the following assumptions:

- The center-of-mass velocity of the sphere is still equal in magnitude but opposite in direction immediately after the collision.
- The sphere is always on the horizontal plane during the collision.
- Friction between sphere and horizontal plane can be neglected during the collision.

Find the time t that it takes from collision to rolling without slipping again. (**Hint:** Use the angular version of impulse-momentum theorem to analyze the angular velocity immediately after collision). **(3 pts)**

Solution: Still, Set into the page as the positive direction for angular velocity, and to the right as the positive direction for velocity. Similar to impulse-momentum theorem, the angular version says $\int \tau dt = \Delta L$. Assume the average normal force from the wall during collision is N, ω is negative right after collision, while collision time is Δt. We have then

$$\text{Impulse-momentum theorem: } \int_0^{\Delta t} N dt = 2mv \qquad \textbf{(0.5 pt)}$$

$$\text{Angular impulse-momentum theorem: } \int_0^{\Delta t} \mu NR dt = \frac{2}{5}mR^2(\omega - (-v/R) \qquad \textbf{(0.5 pt)}$$

$$2\mu Rmv = \frac{2}{5}mR^2(\omega + v/R) \qquad \textbf{(0.5 pt)}$$

$$\therefore \text{Right after collision, angular velocity } \omega = (5\mu - 1)\frac{v}{R} \qquad \textbf{(0.5 pt)}$$

Since $\mu < 0.2$, $\omega < 0$, meaning it is still out of the page. The rest follow the same method as b).

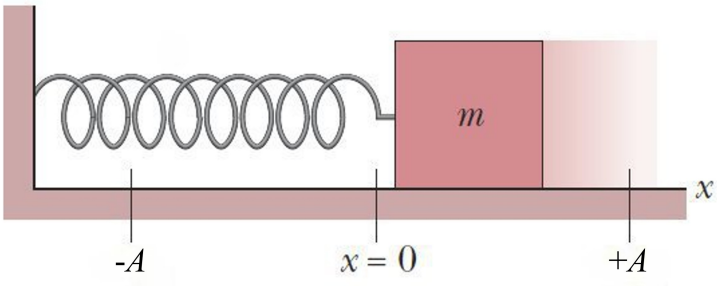
$$\text{Rolling without slipping: } v_f = \omega_f R$$

$$v - \mu gt = (5\mu - 1)v + \frac{5}{2}\mu gt \qquad \textbf{(0.5 pt)}$$

$$t = \frac{2}{7}\frac{2 - 5\mu}{\mu}\frac{v}{g} = \boxed{\frac{4 - 10\mu}{7\mu}\sqrt{\frac{h}{g}}} \qquad \textbf{(0.5 pt)}$$

It takes less time to achieve rolling without slipping after collision if wall has friction.

12. The butterfly effect was first identified by American meteorologist Edward Norton Lorenz in 1961. While running a computer simulation of atmospheric flow, Lorenz restarted a calculation but truncated the input data after the third decimal place, like rounding a number from 0.506127 to 0.506, a seemingly negligible change. To his surprise, the new result diverged completely from the original. This became known as the butterfly effect: "*A butterfly flapping its wings in Brazil might trigger a tornado in Texas.*" It reveals how sensitive nonlinear systems are to initial value problems.



Today, we model this idea with a simple mechanical system:

A mass m is attached to a "softening" spring on a frictionless surface, as shown in the figure above. A is the initial displacement away from $x = 0$. The spring force is

$$F(x) = -kx + \varepsilon x^3$$

where $k > 0$, ε is very small. The x^3 term makes the spring weaker at large x, causing "softening". **(10 pts)**

(a) For a linear spring, $F = -\kappa x$, the spring constant κ is constant. For this nonlinear spring, the restoring force varies with position, its "stiffness" changes can be described using the local spring constant $k_{\text{local}}(x) = -dF/dx$. Find $k_{\text{local}}(x)$ in terms of x, k and ε. Then find the positions where $k_{\text{local}} = 0$. **(1 pt)**

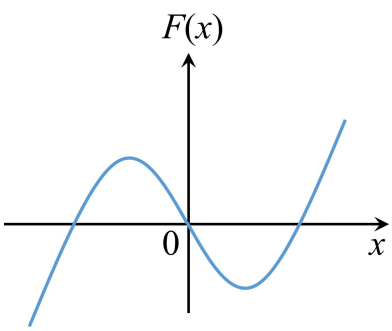
Solution:

$$k_{\text{local}}(x) = -\frac{dF}{dx} = -\frac{d}{dx}(-kx + \varepsilon x^3) = \boxed{k - 3\varepsilon x^2} \qquad \textbf{(0.5 pt)}$$

$$k_{\text{local}}(x) = 0 \Rightarrow \boxed{x = \pm\sqrt{\frac{k}{3\varepsilon}}} \qquad \textbf{(0.5 pt)}$$

(b) Sketch the general shape of force $F(x)$ versus x. (**Hint:** Sketching the shape is OK, no need to mark any number.) **(1 pt)**

Solution:



At $x = 0$, $F = 0$. **(0.5 pt)**

General shape is correct. **(0.5 pt)**

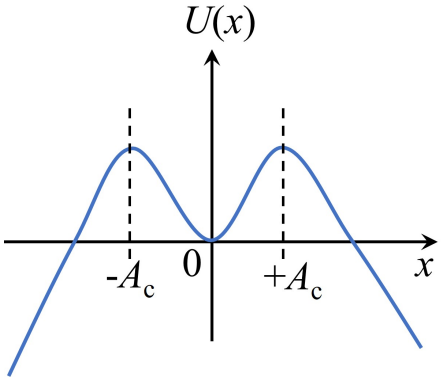
(c) Calculate the system's potential energy $U(x)$. **(1 pt)**

Solution: The potential energy is defined as the negative work done by the spring force:

$$U(x) = -\int_0^x F(x')\,dx' \qquad \textbf{(0.5 pt)}$$
$$= -\int_0^x (-kx' + \varepsilon x'^3)\,dx' = \int_0^x (kx' - \varepsilon x'^3)\,dx'$$
$$= \left[\frac{1}{2}kx'^2 - \frac{1}{4}\varepsilon x'^4\right]_0^x = \boxed{\frac{1}{2}kx^2 - \frac{1}{4}\varepsilon x^4} \qquad \textbf{(0.5 pt)}$$

(d) Sketch the general shape of potential energy $U(x)$ versus x. (**Hint:** Use your answer in (c). Sketching the shape is OK, no need to mark any number.) **(2 pts)**

Solution:



At $x = 0$, $U = 0$, local minimum **(0.5 pt)**

At $x = \pm A_c$, U reaches maximum **(1 pt)**

As $x \to \pm\infty$, $U \to -\infty$ **(0.5 pt)**

Note for grading:

Sketching the general shape is OK, no need to mark $\pm A_c$.

(e) There exists a critical amplitude A_c that the mass can oscillate back and forth. If the amplitude is larger than A_c, the mass flies away and never come back. Find the formula for A_c in terms of k and ε. **(2 pts)**

Solution:

The mass escapes if its total energy exceeds the potential energy barrier at $x = \pm A_c$.

Let $\dfrac{dU}{dx} = kx - \varepsilon x^3 = x(k - \varepsilon x^2) = 0$ **(0.5 pt)** $\Rightarrow$ $x = 0, \pm\sqrt{\dfrac{k}{\varepsilon}}$ **(0.5 pt)**

Check $\dfrac{d^2U}{dx^2} = k - 3\varepsilon x^2$

$\Rightarrow$ At $x = \pm\sqrt{\dfrac{k}{\varepsilon}}$: $\dfrac{d^2U}{dx^2} = -2k < 0$ $\Rightarrow$ local maximum **(0.5 pt)**

So $\boxed{A_c = \sqrt{\dfrac{k}{\varepsilon}}}$ **(0.5 pt)**

Note for grading:

(i) Alternative methods yielding correct answers are acceptable.

(ii) If $\dfrac{dU}{dx}$ and $\dfrac{d^2U}{dx^2}$ were computed in part (d), full credit may still be awarded.

(f) Suppose the mass m is released at $t = 0$ at $x(0) = A_c$ with initial velocity $v(0) = 0$. The total mechanical energy is conserved $E = U(A_c)$. If initial position is changed into $A_c + \delta$, where $0 < \delta \ll A_c$, show that the energy change $\Delta E = E' - E \approx -k\delta^2$. (**Hint:** Expand ΔE to second order in δ.) **(2 pts)**

Solution:

At $x(0) = A_c$, total energy: $E = U(A_c) = \frac{1}{2}kA_c^2 - \frac{1}{4}\varepsilon A_c^4$ **(0.5 pt)**

At $x(0) = A_c + \Delta A$, with $\delta > 0$ small. Expand to second order in δ:

$$E' = U(A_c + \delta) = \frac{1}{2}k(A_c + \delta)^2 - \frac{1}{4}\varepsilon(A_c + \delta)^4 \quad \textbf{(0.5 pt)}$$

$$= \frac{1}{2}k(A_c^2 + 2A_c\delta + \delta^2) - \frac{1}{4}\varepsilon(A_c^4 + 4A_c^3\delta + 6A_c^2\delta^2 + \cdots) \quad \textbf{(0.5 pt)}$$

Substitute $A_c = \sqrt{\frac{k}{\varepsilon}} \Rightarrow \quad \Delta E = E' - E \approx -k\delta^2$ **(0.5 pt)**

(g) Why can this model represent the butterfly effect? **(1 pt)**

Solution:

The initial position determines if the mass oscillates ($x(0) < A_c$) or escapes ($x(0) > A_c$). A tiny change in initial position (e.g., δ) can switch the system from oscillation to escape — this is sensitivity to initial conditions, i.e., the butterfly effect. **(1 pt)**